2.7 Six Digits or Six Hundred

WORKSHEET

The question

STEP II 2023, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Do you notice anything about the two lists? (Look at which fractions appear in both.)

STAGE 2 The error in the first recipe

Ask yourself The question measures the error by $x_n^2 - 2$ rather than $x_n - \sqrt{2}$. Write $x_{n+1}^2 - 2$ as a single fraction in terms of x_n. What do you notice about the numerator?

STAGE 4 The error in the second recipe

Ask yourself Write $y_{n+1} - \sqrt{2}$ as a single fraction. The numerator should remind you of a perfect square.

STAGE 5 Squaring the error

Pause and try Prove by induction that $e_n \le 2r^{2^n}$ for $n \ge 1$, where $r = \frac{\sqrt{2}-1}{2}$. Check the base case carefully.

Try this next

1. **Pell's equation.** Show that if $y_n = p/q$ in lowest terms with $p^2 - 2q^2 = 1$, then $y_{n+1} = \dfrac{p^2 + 2q^2}{2pq}$ also satisfies this equation. Check that $577^2 - 2 \times 408^2 = 1$.

2. **Cube roots.** Write down Newton's method for $f(y) = y^3 - 2$, and show that the new error is $\dfrac{(y_n - \alpha)^2(2y_n + \alpha)}{3y_n^2}$, where $\alpha = \sqrt[3]{2}$. Deduce that if $y_n > \alpha$ then the new error is at most $\frac{1}{\alpha}$ times the square of the old error. (*Hint:* use $\alpha^3 = 2$.)

3. **Division without dividing.** Show that Newton's method for $f(y) = \frac{1}{y} - a$ gives $y_{n+1} = y_n(2 - ay_n)$, which needs no division. If $e_n = 1 - ay_n$, show that $e_{n+1} = e_n^2$. (Computer processors use iterations like this to divide.)

4. **When Newton fails.** Apply Newton's method to $f(y) = y^{1/3}$, whose only root is 0. Show that each step doubles the distance from the root, whatever you start with. Which hypothesis of the error formula fails?

Further reading

- **Endre Süli & David Mayers, *An Introduction to Numerical Analysis*** (Cambridge University Press). The standard undergraduate text; Chapter 1 covers fixed-point iteration and Newton's method.
- **Nick Higham (ed.), *The Princeton Companion to Applied Mathematics*** (Princeton University Press). Short articles on Newton's method and numerical analysis, written by experts for a wide audience.
- **David Fowler & Eleanor Robson, "Square root approximations in Old Babylonian mathematics: YBC 7289 in context"**, *Historia Mathematica* 25 (1998).

The full guided solution, and what the question leads to at university, is in Section 2.7:
https://beyondstep.keithmcnulty.org/ch02/newton-speed.html